PROPOSED COURSE SYLLABUS

COMPUTATIONAL SCIENCE OF MATERIALS

Instructor: Eklavya Thareja | *Office:* TBD | *Classroom:* TBD

Email: eklavya.thareja@gmail.com

Course Level: Upper-level undergraduate/graduate

Prerequisites: Quantum Mechanics and Introduction to Programming

Tools: Python, Quantum ESPRESSO/VASP, Wannier90

Text: R. M. Martin, Electronic Structure: Basic Theory and Practical Methods, 2nd ed.

COURSE DESCRIPTION

Modern materials science arises from applying quantum-mechanical methods to large systems. Combining these methods with computational approaches gives them extraordinary predictive power. This course introduces computational methods such as density functional theory (DFT), Wannier functions, and Python-based analysis that help us understand materials. Rather than learning these as isolated techniques, we will use them to investigate physical questions about real materials: What determines whether a material is metallic or insulating? How does crystal structure influence electronic behavior? When can a first-principles calculation be trusted? Can a complicated electronic structure be reduced to a simpler physical model?

Students will begin by learning the basic theory behind studying many-electron systems (materials) using single-electron methods. They will learn how these approximations explain much of materials' behavior by performing first-principles calculations using DFT and writing programs to analyze the generated data using Python-based tools. We will examine structural, electronic, and magnetic properties; learn how to assess convergence and limitations; and use Wannier90 to construct effective tight-binding models. We will combine these techniques to calculate quantities that are staples of recent publications, such as Berry curvature, the Chern number, and other topological invariants.

The course culminates in an independent computational research project. The goal is not simply to learn how to run calculations, but to develop the ability to decide what to calculate, why to calculate it, whether the result is trustworthy, and what it means physically.

COURSE OBJECTIVE

By the end of the course, students will be able to:

- Formulate physical questions, develop computational models, and investigate them independently.
- Explain the foundations, approximations, and limitations of density functional theory.
- Perform, analyze, and validate DFT calculations, including convergence testing and interpretation of electronic structures.
- Use Python for physical modeling, data analysis, visualization, and computational automation.
- Construct and interpret Wannier-based tight-binding models, explore selected Berry and topological phenomena, and communicate results as evidence-based scientific arguments.

CLASS SCHEDULE

WEEK	DRIVING QUESTION	TOPICS	READING
1	When can we treat many interacting electrons as independent?	Interacting Electrons, Hartree-Fock Approx	Ch. 3
2	How does crystal structure shape electronic behavior?	Crystal/reciprocal space, Bloch states, Brillouin zones	Ch. 4
3	Can we predict a material from first principles?	Many-electron problem, DFT, Kohn–Sham theory	Chs. 6–7.
4	Can we predict the structure of a material?	Plane waves, pseudopotentials, Quantum ESPRESSO	Chs. 11–13
5	How do we know a calculation is trustworthy?	Convergence, k-points, cutoffs, numerical evidence	
6	Can a converged calculation still be wrong?	Approximations, DFT limitations, DFT+U, magnetism, SOC	
7	What does electronic structure tell us?	Bands, DOS, metals, semiconductors, insulators	Chs. 4 & 13
8	What happens when our simplest model fails?	Case study / midterm DFT investigation	
9	Can a complicated band structure be reduced to a simple model?	Tight binding, localized orbitals, Python	Ch. 14-15
10	Can we build an effective model from DFT?	Wannier functions, Wannier90, disentanglement	Ch. 23

WEEK	DRIVING QUESTION	TOPICS	READING
11	What can a Wannier Hamiltonian tell us?	Hopping, interpolation, orbital character, Fermi surfaces	
12	Can quantum-state geometry produce physical effects?	Berry phase, Berry curvature, topology	Chs. 24–26
13	Can electronic structure predict an observable?	SOC, anomalous Hall effect, Wannier interpolation	
14	Can I answer a question nobody gave me?	Independent research projects	

ASSESSMENT

Computational Explorations — 40% of total

Short, low-stakes activities using Python, DFT, and Wannierization. Students will often make predictions before running calculations and reflect on whether the results confirm or contradict their expectations.

Conceptual Problem Sets — 25% of total

Problems focused on the physical ideas underlying DFT, electronic structure, tight binding, Wannier functions, and Berry physics.

Independent Research Project — 35% of total

The independent project evolves alongside the course curriculum. Students select a material system early in the term and progressively apply new computational techniques, from basic DFT to Wannier representations, as they are introduced in class.

- Week 3: Project Proposal & Material Selection (2.5% of total)

  Students select a material system from a curated project menu (or propose an equivalent system) and define a driving physical question focusing on crystal structure, band gaps, or chemical substitution.

- Week 5: Computational Evidence Memo (5% of total)

  Students establish numerical convergence (k-points, energy cutoffs) and run initial structural relaxations on their chosen material.

- Week 8: Baseline Electronic Structure Check (2.5% of total)

  Submission of initial DFT band structures and density of states (DOS) to confirm ground-state properties.

- Week 11: Model Construction & Preliminary Results (5% of total)

  Students apply second-half course concepts (tight-binding fitting, Wannier disentanglement, or Berry curvature) to their existing DFT datasets.

- Week 14: Final Deliverables (20% of total)

  1. Written Report (15% of total): 3–5 page journal-style manuscript synthesizing DFT baseline, model construction, and physical interpretation.
  2. Lightning Presentation (5% of total): 5-minute presentation of key findings.

TEACHING AND LEARNING

This course treats computation as a core method of physical inquiry, not merely a technical skill. We begin with concrete physical phenomena, developing formal concepts as needed to explain what we observe.

Programming is introduced incrementally with fading scaffolding: early activities provide direct structure, while later investigations require independent methodological decisions.

Because collaboration is essential to computational science, students will work together to discuss predictions, debug code, and interpret results, while completing all submitted work individually.

Finally, computational failure is treated as an essential learning tool. Students will practice diagnosing errors, testing assumptions, and iteratively refining their models.

AI AND COMPUTATIONAL TOOLS

AI tools may be used as learning and computational assistants. Appropriate uses include explaining unfamiliar Python syntax, debugging code, interpreting error messages, and suggesting approaches to data analysis.

Students remain responsible for understanding and validating anything they submit. In particular, students should be able to explain:

- What does this calculation do?
- Why is this method appropriate?
- What assumptions are being made?
- How do I know the result is reliable?
- Does the result make physical sense?

The course therefore emphasizes skills that become increasingly important in an AI-enabled scientific environment: asking meaningful questions, evaluating evidence, exercising physical judgment, and communicating scientific reasoning.